



1st gen ROG Phone gets Pie

ASUS has finally rolled out Android 9 Pie for the first gen ROG Phone which first came with 8.1 Oreo. <https://dgit.in/dec19-17>



S. Korea suspends coal plants

Within the next three months, South Korea will suspend 25 percent of its coal-based power plants. <https://dgit.in/dec19-18>

FASCINATING TECH ON CRUISE SHIPS



Take a voyage across the technology on cruise ships and their future

Dhinoj Dings | feedback@digit.in

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f all the different holiday experiences you could have, going on a cruise ship occupies a special

place. Aside from the glitz and comfort cruises offer, there is also the allure of getting away from the maddening crowd on land, to skim across the silvery oceanic waters. But none of these would be possible without the technology that backs cruise ships. So, let's sail across the technologies that make dream-like holidays a reality.

TECH ON CRUISE SHIPS

The building and maintenance of cruise ships make use of some formidable science and technology. Here, we take a look at them.

Saving energy while being docked

The main engines and generators in a ship produce way more energy than required when the ship is docked. The

bulk of the energy generated in a ship goes into moving it on the waters. For instance, about eighty five percent of energy from a diesel electric plant is used to power the ship's propeller. Typically, separate, smaller generators create the power for purposes other than propulsion, like keeping the lights and air conditioning on. However, it's not unheard of for energy to be derived from the main engine to power such



functions while the ship is docked. The downside of this is the increase of emissions. To tackle this problem, it's becoming common today to fit ports and ships with systems so that ships could be plugged into an electric grid on shore when they are docked.

Environment-friendly cruises

In the age of global warming, concerns about the negative impact of emissions are common. This is true in the cruise industry as well. To minimise adverse environmental impact, many ships

these days are powered using Liquefied Natural Gas or LNG. The original engines of these ships are altered so that they can burn natural gas, which is converted and stored in its liquid form. The LNG is stored in insulated tanks, at 260 degrees temperature below zero. This brings down the emission of carbon dioxide by 20 percent and nitrogen oxides by around 80 percent. Another environmentally friendly technology in cruise ships is the hybrid propulsion plant. Following a similar principle used in hybrid cars, these plants use two kinds of power systems- battery packs and diesel engines. The batteries provide emission-free voyage. However, the battery capacity leaves much to be desired at present. As capacity improves, the reliance on diesel engines will come down even more. Attaching small air bubbles under the ship's hull is another way to make cruise ships environment-friendly. These bubbles bring down the surface area of the hull that comes in touch with the sea. This reduces the friction which slows the ship down which translates to better fuel efficiency by up to 10 percent.

The power source

Lugging around mammoth cruise ships is no small feat. Therefore, the engine room which powers the ship is rarely small. In fact, engine rooms are